A model-based tracking method for measuring 3D dynamic joint motion using an alternating biplane x-ray imaging system

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Purposes: To propose a new model-based tracking method for measuring three-dimensional (3D) dynamic joint kinematics using a clinical alternating biplane x-ray imaging system; and to quantify *in vitro* its errors in measuring ankle and knee motions at different motion speeds.
Methods: A new model-based tracking method based on motion component partition and interpolation (MCPI) was developed for measuring 3D dynamic joint kinematics based on a clinical alternating biplane x-ray imaging system. Two detectors of the biplane imaging system placed perpendicular to each other were operated to collect alternating fluoroscopic images of the targeted joint during tasks. The CT data of the joint were also acquired for the reconstruction of volumetric and surface models of each of the associated bones. The CT-based models of the bones were first registered to the alternating images using a model-to-single-plane fluoroscopic image registration method, and the resulting bone poses were then refined using a two-level optimization with motion component partition and model vertex trajectory interpolation. The MCPI method was evaluated *in vitro* for measurement errors for an ankle and a knee specimen moving at different speeds against a standard reference provided by a highly accurate motion capture system. The positional and rotational errors of the measured bone poses were quantified in terms of the bias, precision, and root-mean-squared errors (RMSE), as well as the mean target registration error (mTRE), a final mTRE less than 2.5 mm indicating a successful registration.
Results: The new method was found to have RMSE of bone pose measurements of less than 0.18 mm for translations and 0.72° for rotations for the ankle, and 0.33 mm and 0.74° for the knee with a high successful registration rate (>97%), and did not appear to be affected by joint motion speeds. Given the same alternating fluoroscopic images, the MCPI method outperformed the typical biplane analysis method assuming zero time offset between the two fluoroscopic views. The differences in performance between the methods were increased with increased joint motion speed. With the accurate bone pose data, the new method enabled talocrural, subtalar, and tibiofemoral kinematics measurements with submillimeter and subdegree accuracy, except for an RMSE of 1.04° for the internal/external rotation of the talocrural joint.
Conclusions: A new model-based tracking method based on MCPI has been developed for measuring dynamic joint motions using an alternating biplane x-ray imaging system widely available in medical centers. The MCPI method has been demonstrated *in vitro* to be highly accurate in determining the 3D kinematics of the bones of both the ankle joint complex and the knee. The current results suggest that the MCPI method would be an effective approach for measuring *in vivo* 3D kinematics of dynamic joint motion in a clinical setting equipped with an alternating biplane x-ray imaging system. © *2018 American Association of Physicists in Medicine* [https://doi.org/10.1002/mp.13042]

Key words: ankle, fluoroscopy, knee, model-based tracking, registration

Nomenclature

ACS	anatomical coordinate system
AJC	ankle joint complex
AL	anterior-lateral
DOF	degree-of-freedom
DRR	digitally reconstructed radiograph
FCS	fluoroscopy coordinate system
GCS	global coordinate system
GD	gradient difference
MCPI	motion component partition and interpolation
mTRE	mean target registration error
MVD	motion vector difference
PBA	pseudo biplane analysis
PL	posterior-lateral
RMSE	root-mean-square error

1. INTRODUCTION

Investigation of the biomechanics of the skeleton and the surrounding soft tissues is crucial for a better understanding of the functions and the mechanics of joints in normal, pathological, and postoperation conditions.[1–3] Accurate measurement of *in vivo*, three-dimensional (3D) kinematics of bones and joints could serve as a basis for evaluating functional changes owing to joint disorders and treatments,[4–6] and for deriving soft tissue deformations and forces,[7–9] contributing to the research of musculoskeletal biomechanics.

Skin marker-based stereophotogrammetry has been a major technique for non-invasive, *in vivo* measurement of skeletal motions during the past few decades.[10,11] However, the measurements are subject to errors called "soft tissue artifacts" as a result of undesired motions between the markers and the underlying bones, and are difficult to remove.[12] On the other hand, it is also difficult to identify appropriate locations for placing at least three noncollinear markers on small bones, such as the talus in the ankle and the vertebrae in the spine, limiting their applications in joints formed by small bones. Mathematical approaches have been developed to deal with the artifacts,[13,14] but validation of these methods requires more accurate, independent measurement techniques. The current accuracy of skin marker-based stereophotogrammetry is also limited in the reconstruction of detailed surface kinematics between articulating bones surfaces.

Medical imaging, such as magnetic resonance imaging (MRI) and computed tomography (CT), provides a potentially useful alternative for noninvasive, *in vivo* measurement of 3D skeletal, and thus joint, poses and their surface kinematics, whether large or small. These imaging methods have been used to measure static skeletal and joint poses.[15–18] For dynamic measurements, the development of dynamic MRI technologies for continuous monitoring of planar joint motions[19,20] provided a basis for further development of various approaches for 3D measurements of joint kinematics.[21–23] However, methods based on dynamic MRI imaging are often limited by the restricted volume of the bore of the MRI scanner, in which subjects can only perform isolated joint motions under nonupright, nonweight-bearing conditions.

Model-based tracking techniques using both dynamic x-ray and CT (or MRI) imaging have been shown to provide an effective and less-invasive way to measure directly the 3D kinematics of various joints during upright, weight-bearing functional activities.[24–27] Among the techniques, single-plane fluoroscopy analysis has been used for measuring 3D joint kinematics in normal,[28,29] pathological,[4,30,31] and replaced joints.[32–34] However, the measured skeletal translations normal to the image plane have been found to be substantially less accurate than those for the other components.[35,36] Biplane dynamic x ray has been used to address the issue, as the simultaneous incorporation of an additional x-ray image plane enables a better determination of all six degrees-of-freedom (6-DOF) of individual bones. Such methods have been validated *in vitro* for their accuracy in measurements of kinematics of various joints,[24–26,36–39] and have also been applied to measure rigid-body and intra-articular surface kinematics in various joints during quasistatic[40,41] and dynamic activities.[42–45]

By definition, biplane dynamic x ray-based methods require synchronized imaging by the two x-ray units, which can be achieved only on custom-built biplane systems or by integrating two single-plane C-arm fluoroscopes.[25,37,39,43,46] This may limit the application of these methods in clinical settings because clinical biplane x-ray image systems normally employ an alternating generation/acquisition scheme (i.e., nonsynchronized) in order to minimize the cross-scatter effects. The time offset between the two alternating sequential exposures is constant and may be acceptable for measurements of static bone poses or slow joint motions.[27] However, for fast joint movements the effects of the time offset on the measured skeletal and joint kinematics using the existing methods may lead to errors that should not be overlooked. Therefore, quantifying these effects and developing a new method to reduce such effects are critical for future use of clinical biplane x ray imaging systems in 3D skeletal and joint kinematics measurements.

The purposes of the study were to (a) propose a new model-based tracking method for measuring 3D dynamic joint kinematics using a clinical alternating biplane x-ray imaging system; (b) quantify via *in vitro* validation experiments the effects of joint motion speed on the errors in the kinematics of two commonly studied joints, namely the ankle joint complex (AJC) and the knee joint, obtained from alternating x-ray images using biplane model-based tracking techniques assuming zero time offset (here called pseudo biplane analysis, PBA); (c) evaluate the performance of the proposed new method compared to the PBA.

2. MATERIALS AND METHODS

2.A. The new model-based tracking method

2.A.1. Overview

A new model-based tracking method with motion component partition and interpolation (MCPI) for measuring 3D

dynamic joint kinematics based on a clinical alternating biplane x-ray imaging system was developed. This method requires fluoroscopic images of the targeted joint during tasks from the alternating biplane x-ray imaging system, and the CT data of the joint for reconstructing volumetric and surface models of each of the associated bones. The fluoroscopy images and CT-based models are registered following a two-stage 2D/3D image registration procedure (Fig. 1). At the first stage, a model-to-single-plane fluoroscopic image registration method is applied to give a quick estimate of the bone poses.[47] The bone poses are then refined using a two-level optimization with motion component partition and model vertex trajectory interpolation at the second stage. The two-stage procedure can be applied to the complete image sequence to obtain accurate bone poses for subsequent kinematics analysis. The implementation of the new method in the current study is described in more details as follows.

2.A.2. Biplane image acquisition

A clinical biplane x-ray imaging system (Allura Xper FD20/20, Philips Medical Systems, Netherlands) consisting of two flat panel detectors and two x-ray sources connected on gantries was used in the current study [Fig. 2(a)]. The two detectors were placed perpendicular to each other around an intersection region where the target joint would be imaged. This enabled the acquisition of a pair of orthogonal fluoroscopic images with 1/2 frame time offset, namely anterior-lateral (AL) view (FOV: 30 × 30 cm) and posterior-lateral (PL) view (FOV: 30 × 38 cm) [Fig. 2(a)]. With the capacity of acquiring up to 30 frames/s per individual detector, and using the alternating x-ray generation/acquisition scheme between the two imaging units, the biplane system could eventually achieve a frame rate of up to 60 frames/s.

For model-based tracking, each x-ray source/detector pair was represented as an ideal perspective projection model whose intrinsic parameters (i.e., x-ray source position and virtual image plane) were determined through a calibration procedure using a transparent calibration object with lead markers at known positions.[28,48] A fluoroscopy coordinate system (*FCS*) was embedded in the virtual image plane of each detector with the x-, y-, and z-axes directed horizontally, vertically and normal to the plane, respectively [Fig. 2(a)]. Extrinsic parameters describing the spatial transformation of the two *FCS* were determined following Kaptein et al.[49] using a purpose-built calibration object embedded with 25 lead markers at known positions in two perpendicular planes. The global coordinate system (*GCS*) was then defined in space in parallel with the *FCS* of the PL view (FCS_{PL}). The calibrated alternating biplane imaging system was then used to image the dynamic motions of the targeted joint, which gave two nonsynchronized sequences of fluoroscopic images from two different views.

2.A.3. Bone model reconstruction

The volumetric image data of the targeted joint were collected with a CT scanner (Optima CT660, GE Healthcare,

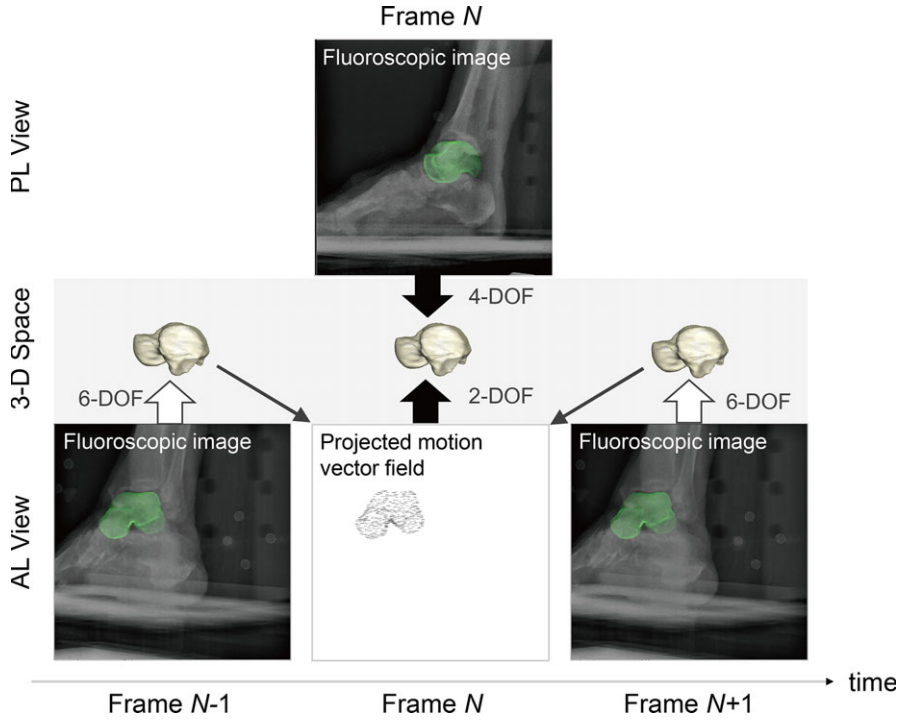


FIG. 1. A schematic representation of the two-stage procedure of the new model-based tracking method. At the first stage, the six degrees-of-freedom (6-DOF) of the bone are determined for each frame using a model-to-single-plane image registration method (here only Frame $N-1$ and $N+1$ are indicated by hollow arrows). At the second stage, taking Frame N as an example, the projected motion vector field is generated from the bone poses of Frames $N-1$ and $N+1$ determined at the first stage (black lines), and used with the fluoroscopic image of Frame N to refine the 6-DOF of the bone using a two-level optimization strategy (solid arrows). (PL: posterior-lateral; AL: anterior-lateral). [Color figure can be viewed at wileyonlinelibrary.com]

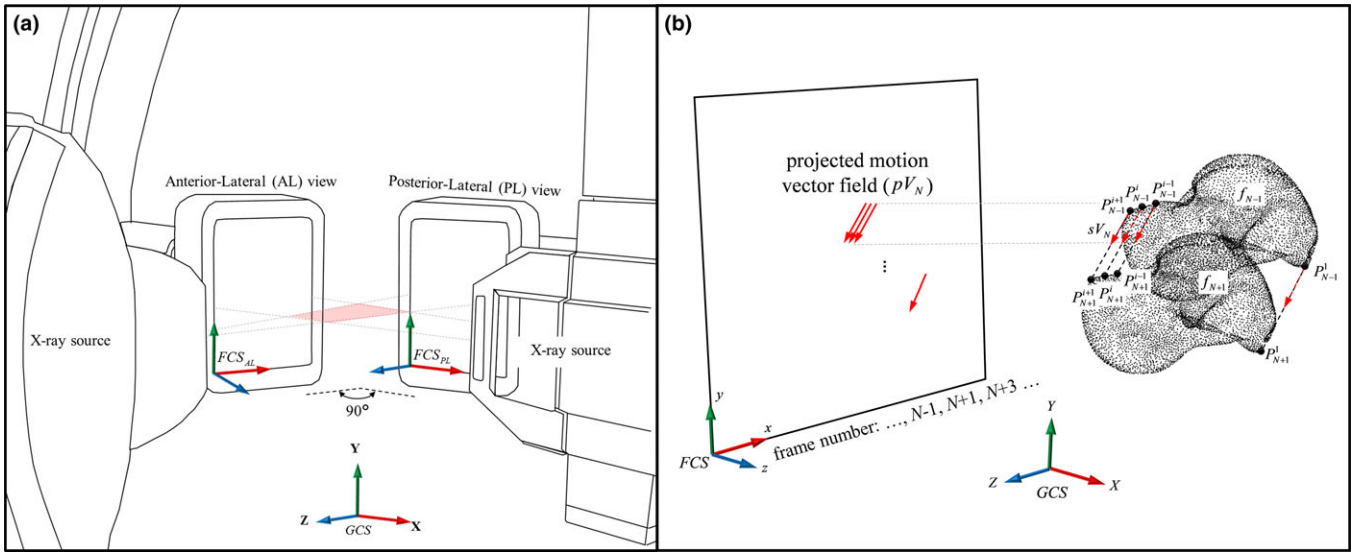


FIG. 2. (a) A clinically available biplane x-ray imaging system consisting of two flat panel detectors and two x-ray sources was used in the current study. Each detector plane was embedded with a fluoroscopy coordinate system (*FCS*) with the x-, y-, and z-axes directed horizontally, vertically, and normal to the plane, respectively. (b) For a detector plane, for example, the AL plane (image frame: . . .$N-1$, $N+1$, . . .), model poses (. . .f_{N-1}, f_{N+1}, . . .) were determined using a model-to-single-plane fluoroscopic image registration method for each image frame, and the projected motion vector field (pV_N) for image frame N was obtained by parallel-projecting onto the AL image plane the spatial motion vector (sV_N), which was linearly interpolated from trajectories of registered model vertices of frames $N-1$ and $N+1$. [Color figure can be viewed at wileyonlinelibrary.com]

Milwaukee, WI, USA), and were then used to reconstruct the computer models of the relevant bones. A seeded region-growing algorithm was implemented to assist manual segmentation of the bone regions for each individual image slice, the thresholds being specified such that the integrity of the bone was maintained while avoiding that the connected set lay on multiple adjacent bones. The volumetric models of individual bones were then obtained from the original CT dataset as the minimum bounding volume containing the complete bone region [Fig. 3(a)]. The voxels that surrounded the bone but which were within the bounding volume were given an intensity value of −1000 [i.e., the Hounsfield unit, (HU) of air]. The Hounsfield unit values of all voxels within the volumetric model were converted to linear attenuation coefficients via a linear transformation formula.[50] Apart from the volumetric models, a marching cubes algorithm[51] was also used to create iso-surfaces of the bones, which were then represented by vertices and faces of triangulations [Fig. 3(b)]. All the necessary image processing for model constructions was performed using in-house-developed software implemented in MATLAB (Mathworks, Inc., Natick, USA).

2.A.4. Model-based tracking

The proposed new MCPI-based tracking method for determining the 3D pose of a bone consisted of a two-stage 2D/3D image registration procedure (Fig. 3):

(1). First stage: A model-to-single-plane fluoroscopic image registration method was used to give a quick estimate of the six degrees-of-freedom (6-DOF) of the bone with respect to GCS.

(2). Second stage: A two-level optimization with model vertex trajectory interpolation and motion component partition strategy was used to refine the 6-DOF of the bone.

First stage image registration: At the first stage, an initial guess of the 6-DOF of the volumetric model of the bone was made and a digitally reconstructed radiograph (DRR) [Fig. 3(c)], $I_{DRR}(u, v)$, was then generated using a ray-tracing with trilinear interpolation method based on the projection model of the x-ray source/detector pair.[48] This was implemented on the GPU to facilitate fast generation of multiple DRRs in parallel. A vectorized genetic algorithm was employed to search for the 6-DOF of the bone whose DRR best matched the corresponding fluoroscopic image, $I_f(u, v)$, according to a similarity measure called *gradient difference* (*GD*),[52] that is, minimizing of the following criterion, f_{GD}, [Fig. 3(d)]:

$$f_{GD} = -\left(\sum_{u,v}\frac{AV}{AV + (I_{dV}(u,v))^2} + \sum_{u,v}\frac{AH}{AH + (I_{dH}(u,v))^2}\right) \quad (1)$$

where

$$I_{dV} = \frac{\partial I_f}{\partial u} - c\frac{\partial I_{DRR}}{\partial u} \quad (2)$$

$$I_{dH} = \frac{\partial I_f}{\partial v} - c\frac{\partial I_{DRR}}{\partial v} \quad (3)$$

and u and v are spatial coordinates. The scaling factor c is determined through an optimization subprocedure. AV and

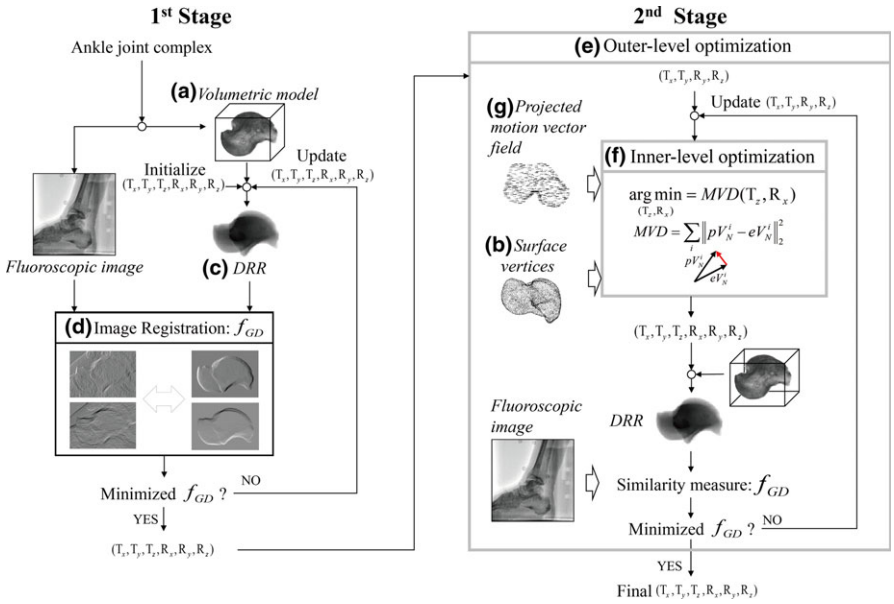


FIG. 3. Flowchart for the proposed model-based tracking method with motion component partition and interpolation. The new method consists of a two-stage image registration procedure. At the first stage, the model bone poses were first obtained for each image frame using a model-to-single-plane fluoroscopy registration method. At the second stage, a two-level optimization strategy incorporating motion component partition and model vertex trajectory interpolation was employed to refine the bone poses. [Color figure can be viewed at wileyonlinelibrary.com]

AH are the variances of the gradient images of I_f. The *GD* was chosen because it has been shown to be a reliable similarity measure for accurate single-plane image registration except for the out-of-plane motion components,[52] and was suitable for incorporating into a parallel computing procedure for fast image registration. The initial guess of the bone pose at the first frame ($N = 1$) was given manually and the initial poses of the bone for subsequent frames ($N > 1$) were defined automatically as the registered poses in the preceding frame. The registration process was performed for all image frames for all bones during the first stage. The registered bone poses were then used as initial bone poses for the subsequent analysis at the second stage. Note that the registration of the first and the last image frames did not involve the second stage image registration as they did not have either the preceding or the subsequent neighboring frames. Therefore, with a robust genetic algorithm, as long as the registration of the first frame at the first stage image is successful, the initial pose of the first frame has no or minimal effect (max deviation <0.5 mm) on the performance of the registration of the subsequent image frames.

Second stage image registration: At the second stage, for an image at frame *N*, acquired either from the PL or AL view, the initial poses of the bones (i.e., registered poses at first stage) were firstly described with respect to the corresponding *FCS* (i.e., FCS_{PL} or FCS_{AL}). The 6-DOF of the bone was then determined via a two-level optimization procedure such that the similarity measure between the DRR of the bone model and the respective fluoroscopic image was minimized. The outer level optimization took four mainly in-plane DOFs of the bone as design variables while the inner level optimization took the remaining two out-of-plane DOFs as design variables. For each iteration of the outer level optimization, an inner level optimization subproblem is solved. This two-level strategy was designed such that the two out-of-plane DOFs to which the similarity measure was less sensitive were left out during the outer level optimization in order to reduce the number of design variables, and only fine-tuned in the inner level optimization subproblem under a different objective function.

The outer level optimization problem took the in-plane translations (T_x and T_y), in-plane rotation (R_z), as well as out-of-plane rotation along the vertical axis (R_y) as design variables and the similarity measure f_{GD} as the cost function [Fig. 3(e)]. An iterative procedure with the Nelder–Mead simplex method[53] was used to find the local minimum of f_{GD}. At each iteration of the outer level optimization, the out-of-plane translation (T_z) and out-of-plane rotation about the horizontal axis (R_x) were determined via an inner level optimization subproblem with T_z and R_x as design variables while keeping the values of the remaining four DOFs fixed (i.e., T_x, T_y, R_z, and R_y). For frame *N*, the inner level optimization subproblem [Fig. 3(f)] was formulated based on the trajectories of the model surface vertices in space from frame $N - 1$ to $N + 1$ [Fig. 2(b)], which were obtained from the corresponding poses that were determined at the first stage image registration (Fig. 1). Given the coordinates of the *i*th vertex at frames $N - 1$ and $N + 1$ (i.e., P^i_{N-1} and P^i_{N+1}), the coordinate of the vertex at frame *N* was defined as the intermediate point P^i_N equidistant from P^i_{N-1} and P^i_{N+1} obtained by using a linear interpolation. The motion vector from P^i_{N-1} to P^i_N was then parallel-projected onto the *x*-*y* plane of the *FCS* at the frame $N - 1$ to obtain a projected motion vector,

pV_N^i [Fig. 2(b)]. The same procedure was repeated for all vertices of the surface model (~1000 points in the current study) to give the projected motion vector field [Fig. 3(g)]. At an iteration of the inner level optimization, given a set of T_z and R_x, and thus the pose of the bone, the vector from P_{N-1}^i to the ith vertex at the current position was also projected onto the same x-y plane, giving the estimated motion vector, eV_N^i. A nonlinear least-squares approach was then used to find the optimum values of T_z and R_x that minimized the motion vector difference (*MVD*) as follows [Fig. 3(f)].

$$MVD = \sum_i \left\| pV_N^i - eV_N^i \right\|_2^2 \tag{4}$$

Once the optimum values of T_z and R_x were obtained, they were used together with the other four DOFs to generate a DRR and the similarity measure value f_{GD} for the next iteration of the outer level optimization.

The proposed model-based tracking method was implemented in MATLAB (Mathworks Inc., Natick, MA, USA) with a graphical user interface, which assisted the user in determining the initial bone poses, and in performing the proposed model-based tracking method. A 64-bit personal computer with an Intel CPU (Core i7-860, 2.8 GHz), an NVIDIA graphics card (GeForce GTX TITAN X) and 16 GB of RAM was used for all numerical computations.

2.B. Performance evaluation

To evaluate the performance of the proposed model-based tracking method, *in vitro* experimental validation studies were conducted on two commonly studied joints, namely the AJC and the knee. The accuracy of the new model-based tracking method and the pseudo biplane analysis (PBA) using nonsynchronized image pairs for measuring the kinematics of the AJC and knee at different motion speeds was evaluated.

2.B.1. Ankle joint complex (AJC)

A fresh-frozen cadaveric specimen of the right foot and ankle complex, ranging from midtibia/fibula to toe, was harvested from a 49-year-old male donor (height: 160 cm; body mass: 60 kg) who received a below-knee amputation for reasons unrelated to abnormalities of the AJC. Marker clusters consisting of an acrylonitrile butadiene styrene base and four 8-mm infrared retroreflective markers each were implanted into the midtibia, the talus, and the calcaneus [Fig. 4(a)].

The specimen construct of the AJC was positioned in a neutral position secured by a plastic jig and CT scanned (Optima CT660, GE Healthcare) at a peak voltage of 120 kVp and a tube current of 180 mA, giving images with a voxel size of 0.623 × 0.623 × 0.625 mm. Under the surveillance of the biplane imaging system and a motion capture system with 14 infrared cameras (VICON 612, Oxford Metrics, Oxford, UK), the AJC specimen, affixed to a stationary bar at the forefoot, was placed at the isocenter of the biplane imaging system, positioned to simulate a standing posture [Fig. 4(a)] and driven manually by moving the proximal end of the tibia for full ranges of dorsi-plantar flexion, valgus–varus, and internal-external rotations, each for three continuous cycles at three cadences, namely 1/3, 1/2, and 1 cycle/s. The source-to-specimen and the specimen-to-detector distances at the static position were approximately 860 mm and 320 mm, respectively. A linear translation trial was also performed with the specimen placed on a treadmill traveling at a constant speed of 1 km/h.

For each of the trials, alternating fluoroscopic images of the specimen with a resolution of 512 × 512 pixels were acquired by the biplane imaging system at 60 Hz, 40–52 kVp, and 5 mA for subsequent model-based tracking analysis. The 3D trajectories of the marker clusters were also measured simultaneously by the motion capture system at 120 Hz and used to determine the standard reference of the bone kinematics.

2.B.2. Knee joint

A fresh-frozen cadaveric specimen of the left knee, from midfemur to midtibia, was harvested from a 59-year-old male donor (height: 165 cm; body mass 63 kg) who received a below-hip amputation for reasons unrelated to abnormalities

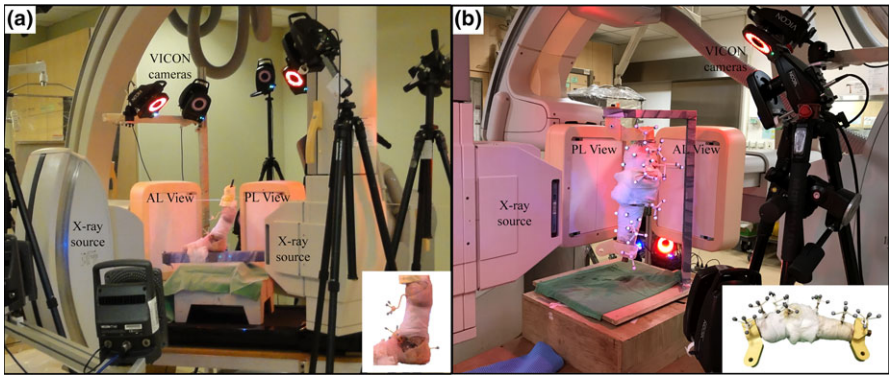


FIG. 4. Setup for the validation experiments for (a) ankle and (b) knee specimens. The ankle specimen was fixed to a stationary bar at the forefoot while the knee specimen was mounted on a testing rig. The infrared cameras were placed around the field of view of the biplane imaging system to capture the coordinates of the marker clusters implanted in the specimens. [Color figure can be viewed at wileyonlinelibrary.com]

of the knee. The proximal end of the femur and distal end of the tibia were rigidly fixed to a radiolucent attachment. Seven marker cluster sets, each with four 14-mm infrared retroreflective markers on a 3D printed polylactic acid frame, four implanted in the femur, and three sets of clusters in the tibia via radiolucent poly-ether-ether-ketone screws, were used to provide the standard reference motions of the femur and tibia [Fig. 4(b)]. The specimen/attachment construct with the marker clusters was then positioned in lateral recumbency and CT scanned at a peak voltage of 120 kVp and a tube current of 180 mA, giving images with a voxel size of $0.709 \times 0.709 \times 0.625$ mm.

The specimen/attachment construct was then mounted on a testing rig with the knee joint at approximately the isocenter of the biplane imaging system, enabling image acquisition of the joint. The source-to-specimen and the specimen-to-detector distances at the static position were approximately 820 mm and 280 mm, respectively. The femoral attachment was initially mounted on the rig via a hinge while the tibial attachment was connected to a plastic bar that was used to rotate the tibia manually, simulating open-chain knee flexion/extension [Fig. 4(b)]. The motion data were collected for three continuous cycles at five different cadences, namely 0.25, 0.375, 0.5, 0.625, and 0.75 cycle/s. The same experimental procedure was repeated while the tibial attachment was mounted on the rig and the femoral attachment was connected to the bar. This inverted configuration of the specimen was used to simulate closed-chain movement of the knee by rotating the femur. The alternating fluoroscopic images of the specimen with a resolution of 512×512 pixels were collected at 60 Hz, 50–59 kVp, and 5 mA for subsequent model-based tracking analysis, while the 3D coordinates of the marker clusters were collected simultaneously using the motion capture system at 120 Hz and were used to determine the standard reference of the bone kinematics.

2.B.3. Standard reference bone poses

For the AJC, the distal tibia, talus, and calcaneus were embedded with anatomical coordinate systems (ACS) defined using anatomical landmarks of the bones.[43] For the knee joint, the ACS of the distal femur and proximal tibia were also defined through an automated procedure based on the anatomical features of the bones.[54] All the ACS had the positive x-axis directed anteriorly, the positive y-axis directed superiorly and the positive z-axis directed to the right.

The marker coordinates of the clusters in the GCS, namely P_{GCS}, were obtained via the spatial transformation between the GCS and the coordinate system of the motion capture system following a previously reported procedure with a custom-made calibration object.[55] The coordinates of the same markers in the corresponding bone ACS were also derived from the CT data, which were then registered to P_{GCS} via a least-squares algorithm,[56] giving the standard reference of the 6-DOF of the bones. The precisions of the marker coordinates for individual bones over the trials were less than 0.17 mm as determined by the standard deviations of intermarker distance differences obtained between CT and the motion capture system. The standard reference of the 3D kinematics of a joint was then defined as the pose of the distal bone relative to the proximal one. The angles of the talocrural and subtalar joints were determined as the rotation of the ACS of a distal bone with respect to that of a proximal bone following a z-x-y Cardanic rotation sequence corresponding to dorsi/plantar flexion, inversion/eversion, and internal/external rotation.[57] The same rotation sequence was also applied to the tibiofemoral joint, giving flexion/extension, adduction/abduction, and internal/external rotation.[58] The joint center translations were defined as the displacements of the origin of the proximal ACS relative to that of the distal ACS.

2.B.4. Pseudo biplane analysis (PBA)

The alternating x-ray images were also analyzed using the PBA method, a typical model-to-biplane fluoroscopic image registration approach.[25,36,39] The PBA performed model-based tracking assuming zero time offset, that is, assuming synchronized acquisition, and ignoring the nonsynchronized nature of the alternating x-ray images. In the current study, the PBA determines the 3D poses of the bones by minimizing the following combined cost function.

$$f_{PBA} = \sum_{j=1}^{2} f_{GD}^{j} \tag{5}$$

where j denotes the number of the fluoroscopic detector.

2.B.5. Error metrics

The proposed MCPI method was used with each of the bone models to determine the corresponding spatial bone poses for the complete sequence of the image frames. For comparing the performance of the MCPI to the PBA method, both analyses were given the same initial bone poses for each image frame. The differences between the measured 6-DOF of the bones and joints, and those of the standard references, were taken as measurement errors. The root-mean-squared errors (RMSE) of the measurements across all image frames were calculated for each trial. The means and standard deviations (SD) of the measurement errors gave the bias and precision, respectively, of the method used. The positional and rotational errors of the measured bone poses were also quantified by the mean target registration error (mTRE) of surface points on the bone model.[59] A registration was defined as successful when the final mTRE was less than a threshold of 2.5 mm (obtained by averaging mTREs of the initial bone poses), which was considered reasonable because the reported accuracy of 2D/3D registrations of bones ranged from 0.88 to 7.8 mm.[25,35,47,60] A similar threshold was also chosen by Kraats et al.[59]

To assess the effects of bone movement velocities on the measurement errors, the velocity index of a trial, V_t, was defined and computed as:

$$V_t = \frac{1}{K-1}\sum_{N=2}^{K}\left(\frac{1}{M}\sum_{i=1}^{M}\left(\left|P_N^i - P_{N-1}^i\right|\Big/\Delta t\right)\right) \quad (6)$$

where M and K represented the number of vertices of the bone model and the number of the image frames of the trial, respectively. The Δt represented the time interval between image frames. The performances of the MCPI and the pseudo biplane analysis in tracking bone motions were also compared at an average movement velocity corresponding to that during the stance phase of typical level walking.

The sensitivity of the performance of the MCPI method to the bone segmentation was evaluated by dilating and eroding the original segmented bone region by one pixel for each CT image slice — an overestimated manual error condition — and the bone pose estimation for a trial from the femoral motion experiment was redone using the MCPI method with each reconstructed bone model. The changes in the measurement errors owing to contour dilation and erosion were then obtained.

3. RESULTS

With the two-stage strategy the new model-based tracking method was shown to have achieved rotational and translational errors of less than 1° and 1 mm, respectively, for bones of both the AJC and the knee (Fig. 5). At the first stage, the method relied on the registration of the bone model with single-plane fluoroscopic images, which gave submillimeter and subdegree errors for in-plane motion components but had greater errors for the out-of-plane motion components (Fig. 5). The two-level optimization at the second stage further reduced the RMSE of out-of-plane translations of the bones (along the z-axis) from 3.37 to 0.18 mm for AJC and from 2.98 to 0.33 mm for the knee (Fig. 5). The RMSE of out-of-plane rotations (about the x-axis) of the bones were also reduced from 0.62° to 0.28° and from 0.65° to 0.41° for AJC and knee, respectively, while the RMSE of the other components were only slightly reduced (Fig. 5).

At the average movement velocity corresponding to that during the stance phase of typical level walking, the MCPI showed smaller errors for the distal tibia of the AJC specimen in the z-axis translation and x-axis rotation when compared to the pseudo biplane analysis (Fig. 6). For the distal femur, both the ranges and the 25th and 75th percentiles of the errors for all the motion components obtained by the MCPI method were smaller than those obtained by the pseudo biplane analysis (Fig. 6).

Effects of movement speeds of the bones on the errors of the tested methods were evaluated in terms of the mTRE with respect to velocity index. The results showed that the mTRE obtained from pseudo biplane analysis tended to grow while the velocity index was above 50 mm/s (Fig. 7). The higher the movement speeds, the greater the differences in mTRE between the methods. Once the velocity indices were greater than 180 mm/s, the mTRE was greater than 2.5 mm for the five tested bones when using the pseudo biplane analysis (Fig. 7). The MCPI method was shown to be effectively integrated with the non-synchronized images, which gave mTRE of less than 1.0 mm even under high speed movement conditions (Fig. 7).

For the AJC, over all the trials, the successful registration rate for the MCPI was higher than 99.5%, with absolute values of the bias less than 0.09 mm for translations and 0.23° for rotations. The means of the precision were less than 0.19 mm for translations and 0.89° for rotations (Table I). The overall RMSE over the ankle bones were 0.16 ± 0.10 mm, 0.13 ± 0.10 mm and 0.18 ± 0.11 mm for x-, y-, and z-axis translations, respectively. The corresponding values for x-, y-, and z-axis rotations were 0.28 ± 0.15°, 0.72 ± 0.24°, and 0.34 ± 15°, respectively (Table I). For the knee joint, the MCPI gave a successful registration rate of 97.7% for the femur and 98.2% for the proximal tibia (Table I). The RMSE over the femur and proximal tibia were 0.31 ± 0.10 mm, 0.33 ± 0.08 mm, and 0.33 ± 0.10 mm for x-, y-, and z-axis translations, respectively. The corresponding values for x-, y-, and z-axis rotations were 0.41 ± 0.07°, 0.74 ± 0.14°, and 0.49 ± 10°, respectively (Table I). The ensemble mean changes in the measurement errors of the MCPI method owing to the segmentation variations (i.e., contour dilation and erosion) were all within 0.1 mm for translations and 0.2° for rotations, suggesting that the performance of the MCPI method did not appear to be sensitive to contour point variations within two pixels.

The curves of the motion components of the tibiofemoral joint over three cycles of knee flexion/extension obtained by the MCPI method followed closely those of the standard

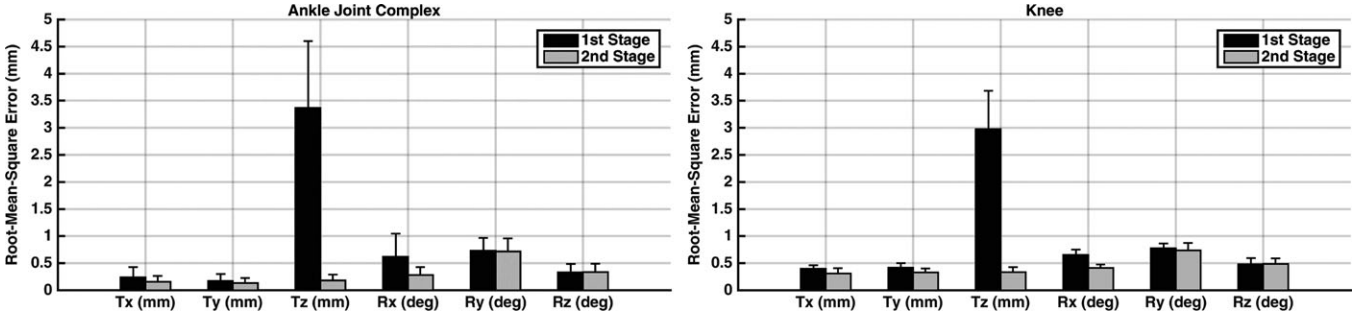


FIG. 5. Means and standard deviations of the root-mean-square errors for each of the motion components of the bones with respect to the corresponding fluoroscopic coordinate system over the tested trials after the first stage process (black bars) and after the second stage process (gray bars) for the ankle joint complex and knee. T_x and T_y, and R_z are in-plane translations and rotations, respectively. R_x and R_y, and T_z are out-of-plane rotations and translations, respectively.

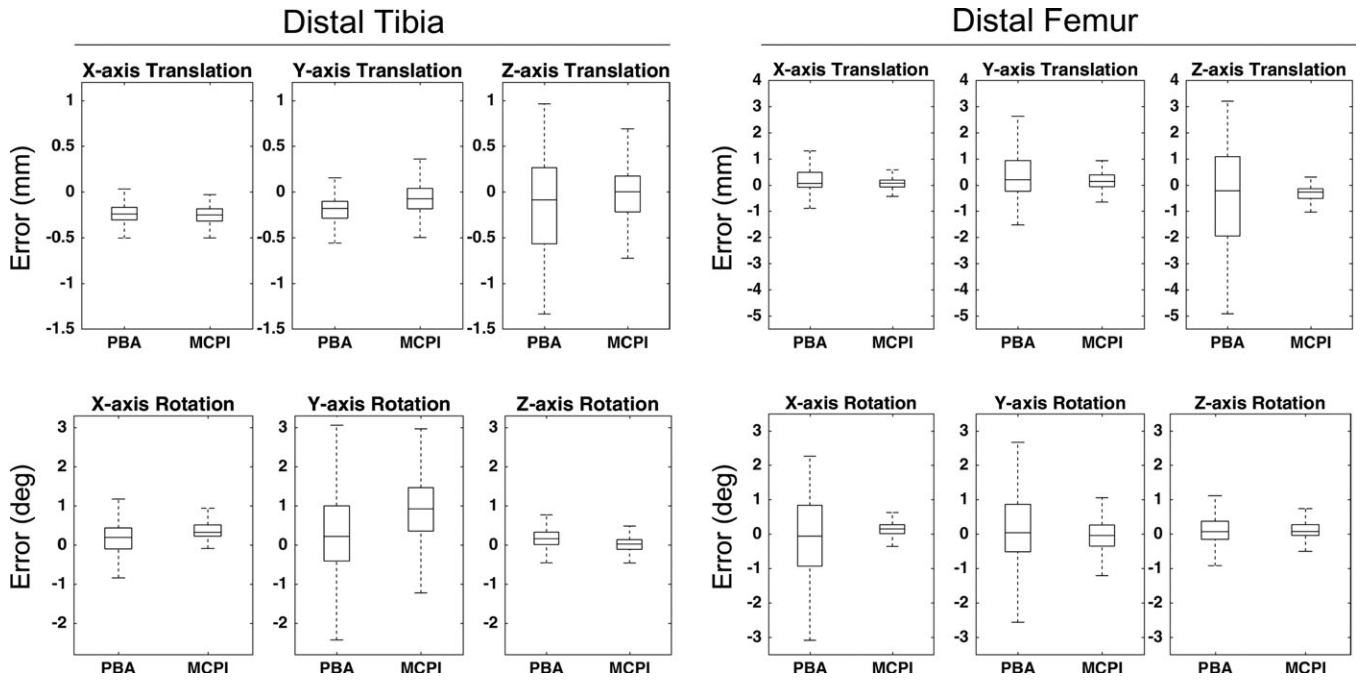


FIG. 6. Error distributions of the distal tibia and distal femur by new model-based tracking with motion component partition and interpolation method (MCPI) and pseudo biplane analysis (PBA) at the average movement velocity corresponding to that during the stance phase of typical level walking. The central lines of the boxes indicate the median values; the top and bottom edges of the boxes indicate the 75th and 25th percentiles of the error distributions; and the ranges of the whiskers indicate the upper and lower extremes.

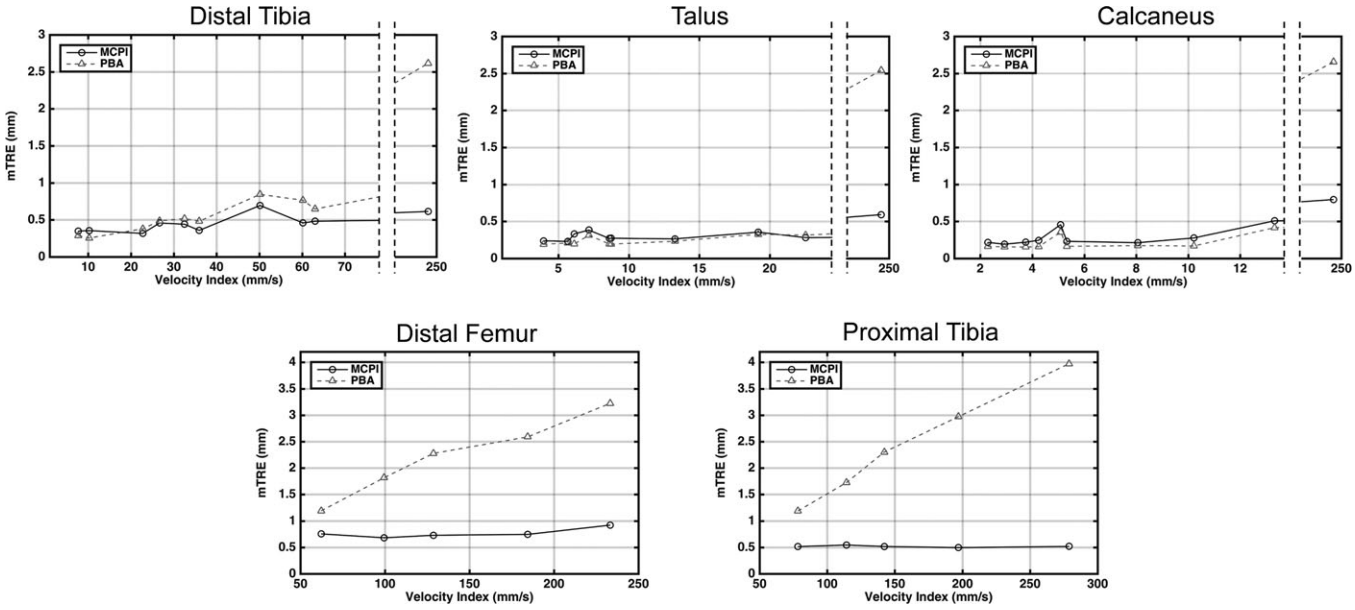


FIG. 7. Final mean target registration errors (mTRE) with respect to the velocity index for the bones of the ankle (distal tibia, talus, and calcaneus) and the knee (distal femur and proximal tibia) using pseudo biplane analysis (PBA, triangle markers) and the new model-based tracking with motion component partition and interpolation method (MCPI, circle markers).

reference joint poses (Fig. 8). In general, the MCPI method showed submillimeter and subdegree accuracy in measuring the kinematics of the three tested joints, except for an RMSE of 1.04° for internal/external rotation (i.e., PD-axis rotation) of the talocrural joint (Table I).

4. DISCUSSION

The study proposed a new model-based tracking method specifically developed for a clinical alternating biplane x-ray imaging system to measure dynamic, 3D kinematics of joints;

TABLE I. Means ± SD of the root-mean-square (RMS) error, bias and precision over all the measurements, successful registration rate (SR rate) and run time using the model-based tracking with motion component partition and interpolation method. The errors are expressed with respect to the fluoroscopy coordinate system of the corresponding image plane for bone pose errors and the joint coordinate system for joint pose errors.

		ΔT_X (mm)	ΔT_Y (mm)	ΔT_Z (mm)	ΔR_X (°)	ΔR_Y (°)	ΔR_Z (°)	Run time (s) stage#1/stage#2
Ankle joint complex								
Distal tibia (SR rate: 99.5%)	RMS	0.17 ± 0.09	0.23 ± 0.12	0.20 ± 0.09	0.18 ± 0.09	0.94 ± 0.08	0.23 ± 0.09	5.1 ± 0.8/15.2 ± 4.9
	Bias	−0.08 ± 0.05	−0.09 ± 0.13	0.04 ± 0.13	0.06 ± 0.05	0.23 ± 0.24	0.06 ± 0.09	
	Precision	0.15 ± 0.09	0.19 ± 0.07	0.17 ± 0.07	0.17 ± 0.08	0.89 ± 0.13	0.21 ± 0.10	
Talus (SR rate: 99.9%)	RMS	0.15 ± 0.09	0.09 ± 0.01	0.19 ± 0.09	0.43 ± 0.12	0.64 ± 0.18	0.50 ± 0.12	3.7 ± 0.2/12.2 ± 7.3
	Bias	−0.04 ± 0.04	−0.04 ± 0.02	0.05 ± 0.12	0.09 ± 0.23	−0.11 ± 0.15	−0.19 ± 0.17	
	Precision	0.15 ± 0.09	0.07 ± 0.01	0.16 ± 0.06	0.36 ± 0.10	0.63 ± 0.16	0.45 ± 0.07	
Calcaneus (SR rate: 100.0%)	RMS	0.15 ± 0.14	0.08 ± 0.02	0.16 ± 0.14	0.22 ± 0.10	0.57 ± 0.24	0.27 ± 0.08	4.3 ± 0.4/12.1 ± 5.8
	Bias	−0.06 ± 0.06	0.05 ± 0.02	−0.06 ± 0.15	−0.08 ± 0.12	−0.23 ± 0.36	0.12 ± 0.04	
	Precision	0.13 ± 0.13	0.06 ± 0.00	0.12 ± 0.07	0.19 ± 0.04	0.45 ± 0.08	0.24 ± 0.08	
Bone overall (SR rate: 99.8%)	RMS	0.16 ± 0.10	0.13 ± 0.10	0.18 ± 0.11	0.28 ± 0.15	0.72 ± 0.24	0.34 ± 0.15	N.A.
	Bias	−0.06 ± 0.05	−0.03 ± 0.10	0.01 ± 0.14	0.02 ± 0.17	−0.04 ± ± 0.32	0.00 ± 0.18	
	Precision	0.14 ± 0.10	0.11 ± 0.07	0.15 ± 0.07	0.24 ± 0.12	0.66 ± 0.22	0.30 ± 0.13	
Talocrural joint	RMS	0.40 ± 0.41	0.25 ± 0.11	0.31 ± 0.09	0.54 ± 0.16	1.04 ± 0.19	0.54 ± 0.16	N.A.
Subtalar joint	RMS	0.27 ± 0.12	0.26 ± 0.08	0.36 ± 0.12	0.51 ± 0.09	0.81 ± 0.18	0.60 ± 0.19	N.A.
Knee								
Distal femur (SR rate: 97.7%)	RMS	0.40 ± 0.04	0.40 ± 0.02	0.42 ± 0.04	0.35 ± 0.04	0.61 ± 0.02	0.41 ± 0.05	13.7 ± 3.1/18.7 ± 5.9
	Bias	−0.11 ± 0.02	0.26 ± 0.02	−0.19 ± 0.01	0.12 ± 0.02	0.11 ± 0.01	−0.03 ± 0.03	
	Precision	0.38 ± 0.04	0.30 ± 0.04	0.37 ± 0.04	0.33 ± 0.04	0.60 ± 0.02	0.41 ± 0.05	
Proximal tibia (SR rate: 98.2%)	RMS	0.22 ± 0.03	0.26 ± 0.01	0.25 ± 0.04	0.47 ± 0.03	0.87 ± 0.03	0.57 ± 0.07	8.8 ± 1.9/15.5 ± 4.9
	Bias	−0.03 ± 0.02	0.05 ± 0.01	−0.03 ± 0.01	0.10 ± 0.06	0.18 ± 0.10	0.08 ± 0.04	
	Precision	0.22 ± 0.03	0.25 ± 0.01	0.25 ± 0.04	0.45 ± 0.02	0.84 ± 0.02	0.56 ± 0.07	
Bone overall (SR rate: 97.9%)	RMS	0.31 ± 0.10	0.33 ± 0.08	0.33 ± 0.10	0.41 ± 0.07	0.74 ± 0.14	0.49 ± 0.10	N.A.
	Bias	−0.07 ± 0.05	0.16 ± 0.11	−0.11 ± 0.08	0.11 ± 0.04	0.14 ± 0.07	0.02 ± 0.07	
	Precision	0.30 ± 0.09	0.28 ± 0.04	0.31 ± 0.08	0.39 ± 0.07	0.72 ± 0.13	0.48 ± 0.10	
Tibiofemoral joint	RMS	0.41 ± 0.02	0.39 ± 0.10	0.44 ± 0.11	0.50 ± 0.05	0.96 ± 0.29	0.26 ± 0.07	N.A.

and to evaluate it quantitatively for its accuracy in measuring kinematics of the AJC and the knee compared with that obtained by pseudo biplane analysis. It appeared that the model-based tracking with MCPI could successfully reduce errors in the out-of-plane components when compared with the model-based tracking using single-plane fluoroscopic images. With the MCPI, no substantial effects of the movement speeds on measurement errors were found. In general, the new model-based tracking method incorporating the MCPI strategy was capable of determining 3D kinematics of the bones during dynamic activities with bias and precision of less than 0.9° and 0.4 mm for rotations and translations, respectively.

Clinical biplane x-ray imaging systems have a great potential for their widespread application in model-based tracking for 3D dynamic joint kinematics measurements, as opposed to custom-made biplane x-ray imaging systems. This is mainly because they are standard equipment in most medical centers, and are thus available for measurements without additional costs for a new system. However, the alternating generation/acquisition scheme in clinical biplane imaging systems may affect the accuracy of the existing model-based tracking methods with which the 3D bone models were best registered to the x-ray images of two views (here referred to as PBA). In the current study, under the movement speed of typical gait the use of PBA was shown to result in considerable errors in bone poses obtained (Fig. 6). The errors were increased with increased lower extremity movement speed (Fig. 7), which may limit further applications in measuring fast body motions.

To bridge the gap, the proposed new method was designed to accommodate the alternating generation/acquisition nature of most clinical biplane x-ray imaging systems. Overall, the proposed MCPI method had a success rate as high as 97%. The image frames that failed during the registration were those with incomplete bone images as a result of overlapping of bones or out-of-detector views, leading to far from correct bone poses at the first stage, erroneous motion vectors at the second stage, and finally, bone poses with mTRE greater than the 2.5 mm threshold. The new MCPI-based tracking method using alternating biplane x-ray image systems gave accuracy values that were well within those obtained by other model-based tracking methods using custom-built synchronized biplane x-ray imaging systems, mostly with submillimeter

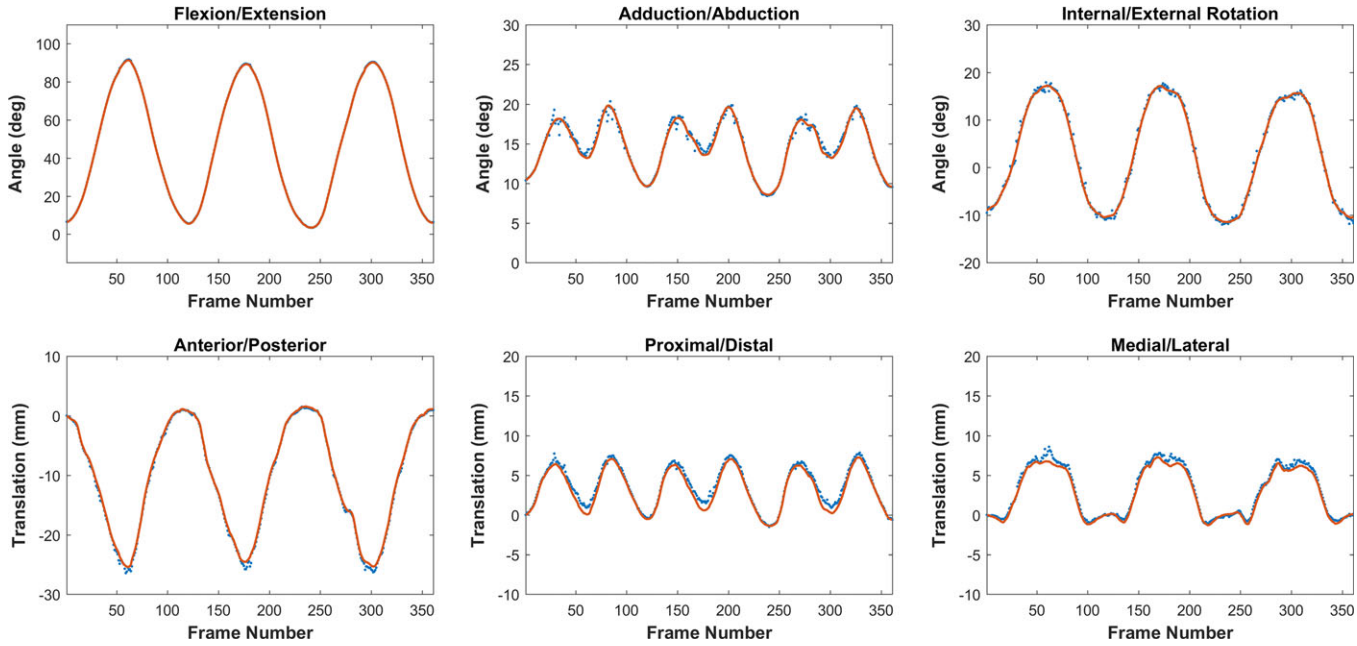


FIG. 8. Motion components of the tibiofemoral joint over three cycles of knee flexion/extension obtained by the proposed model-based tracking method with motion component partition and interpolation (MCPI, dotted curves) and those of the standard reference joint poses (solid curves). [Color figure can be viewed at wileyonlinelibrary.com]

and subdegree accuracy.[24,25,38,61,62] While a direct comparison between these results and the current study may not be feasible considering the different scanners used, specimen status, accuracy representations, and experimental conditions and protocols, the reported values provide a possible range of accuracy for custom-built synchronized biplane x-ray image systems. With alternating biplane systems, the new model-based tracking method showed standard deviations of errors (precision) of less than 0.84° in rotations and 0.38 mm in translations for the knee bones, which were well within the reported ranges of 0.65–0.91°and 0.18–0.7 mm, respectively.[24,25,61] For the AJC, the new method also gave errors comparable to those reported in a previous validation study,[62] the differences in the translational and rotational bias and precision being less than 0.1 mm and 0.2°, respectively. The measurement accuracy for the new method did not appear to be affected by the movement speed within the tested range (Fig. 7). Therefore, the new MCPI-based method is considered to be highly accurate for joint movement measurements. Note that the infrared markers present in the CT and fluoroscopic images were not filtered out during the registration process. This may have negative effects on the measurement accuracy of the registration methods. Therefore, the reported accuracy may be regarded as the upper bound, which theoretically can be reduced further in applications not involving markers.

The new model-based tracking method was based on a two-level optimization scheme, the outer level being the 3D-to-2D single-plane image registration for one image plane while the inner level incorporated spatial geometric information from the other image plane. The outer level optimization ensured accurate determination of the in-plane motion components by best matching the DRR and the corresponding fluoroscopic images, while the inner level optimization aimed to search for a set of out-of-plane motion components such that the projected geometry of the bone best-complied with the projected trajectories provided by the preceding and the subsequent frame of the other view. With a single-plane image registration, the 6-DOF are generally chosen to maximize the similarity between the DRR and the fluoroscopic image. This does not always produce accurate results because factors such as image artifacts, DRR generation, calibration errors, and projection model imperfections all contribute to the outcome. In fact, single-plane image registration often has greater errors in the out-of-plane components because the similarity measure is not sensitive to the out-of-plane movement. By estimating the out-of-plane components of the image plane from the other image plane (i.e., the inner level optimization in the new method), more accurate results can be obtained at the outer level even though the same similarity measure between the DRR and the fluoroscopic image is used.

The proposed MCPI algorithm relied on the projected motion vectors to revise the out-of-plane translations and rotations. The major factors affecting the projected motion vectors included the accuracy of in-plane motions of the bones at adjacent frames and the methods used to derive the motion vectors. The high accuracy of in-plane motion components obtained from the first stage has been reported in the literature[47] and in the current study (Fig. 5). The intermediate point P_N^i between P_{N-1}^i and P_{N+1}^i (Fig. 2) was the key in determining the projected motion vector used in the subsequent image registration procedure, thus accurate determination of these intermediate points is of great importance in the

final measurement accuracy. In the current study, linear interpolation was used to estimate an intermediate point to minimize the computational effort because it required only the information from the two adjacent frames. More sophisticated interpolation techniques, such as higher order interpolating polynomials and spline functions, may be used instead, but information from more image frames would be needed. On the other hand, the current study used parallel projection to generate the projected motion vector field mainly owing to its computational simplicity. Further study would be needed to confirm whether different projection methods (parallel vs perspective) would perform differently at the second stage image registration.

In the current study, the two detectors of the biplane imaging system were configured to be perpendicular to each other to collect images of the joints from *AL* and *PL* perspectives when the tested joints were placed at the isocenter of the biplane imaging system. As a result, the out-of-plane translation and rotation (i.e., T_z and R_x) of one image plane corresponded approximately to the in-plane motion components of the other image plane (i.e., T_x and R_z.), which would be ideal in reducing errors in T_z and R_x components. Although the *y*-axes were parallel to each other for two *FCS* with the current configuration, which was not expected to improve the accuracy, a subdegree accuracy in the *y*-axis rotation, R_y, was obtained with the new method (maximum RMSE: 0.94, Table I). Further studies would be needed to determine whether new configurations with nonparallel y-axes and different positions of the tested specimen within the biplane imaging system would affect the accuracy of the proposed MCPI-based tracking method.

5. CONCLUSIONS

A new model-based tracking method based on MCPI has been developed for measuring dynamic joint motions using an alternating biplane x-ray imaging system widely available in medical centers. The new method has been demonstrated *in vitro* to be highly accurate in determining the 3D kinematics of the bones of both the AJC and the knee with an averaged RMSE of 0.33 mm for translations and 0.74° for rotations. The current results suggest that the new method would be an effective approach for measuring *in vivo*, 3D kinematics of dynamic joint motion in a clinical setting equipped with an alternating biplane x-ray imaging system.

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CONFLICTS OF INTEREST

The authors declare that they have no financial or personal relationships with other people or organizations that could inappropriately influence or bias the work or pose conflicts of interest regarding the research.

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